



MARRI LAXMAN REDDY

INSTITUTE OF TECHNOLOGY & MANAGEMENT

(AN AUTONOMOUS INSTITUTION)

(Approved by AICTE, New Delhi & Affiliated to JNTUH, Hyderabad)

NAAC Accredited Institution with 'A' Grade & Recognized Under Section 2(f) & 12(B) of the UGC act, 1956

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Advanced Data Structures

LAB PROJECT

Autocomplete System using Trie Data Structure

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COMPUTER SCIENCE AND ENGINEERING



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CERTIFICATE

This is to certify that the Lab Project Report entitled “**Autocomplete System using Trie Data Structure**” is being submitted by **Srinivas Rao Tammireddy (257Y1D5805)** in **M.Tech I Year I Semester COMPUTER SCIENCE AND ENGINEERING** is a record bonafide work carried out by him/her. The results embodied in this report have not been submitted to any other university for the award of any degree.

Internal Examiner

Head of the Department

Abstract

An Autocomplete System is a widely used feature in modern applications such as search engines, code editors, and messaging platforms. This project designs and implements a fully functional Autocomplete System using the Trie (Prefix Tree) data structure in JavaScript. The Trie enables extremely efficient prefix-based word retrieval with $O(m)$ time complexity, where m is the length of the prefix, independent of the total number of words stored.

The system supports four core operations: word insertion with frequency weighting, prefix-based autocomplete search using Depth-First Search (DFS), exact-match lookup, and word deletion with bottom-up node pruning to reclaim unused memory. Suggestions returned by the autocomplete engine are ranked in descending order of frequency, enabling priority-based retrieval.

The project also includes an interactive web-based visualiser built with HTML, CSS, and JavaScript that renders the Trie structure dynamically as words are inserted and deleted. The visualiser highlights the active prefix path, distinguishes word-end nodes from intermediate nodes, and displays real-time performance statistics including node count, word count, and search time in milliseconds.

Experimental results demonstrate that prefix search on a Trie with over 25 inserted words completes in under 0.1 ms in the browser environment. The node count remains significantly lower than the worst-case bound due to shared prefix compression, confirming the structural efficiency of the Trie compared to alternative approaches such as hash maps or sorted arrays.

Introduction

Autocomplete, also known as word completion or predictive text, is a feature that suggests possible completions to a user based on a prefix they have typed. It is a fundamental feature in search engines (Google, Bing), integrated development environments (VS Code, IntelliJ), mobile keyboards, and database query tools. The effectiveness of an autocomplete system depends critically on the speed at which matching suggestions can be retrieved and ranked.

A naive implementation might store all words in an array or hash map and scan every entry for each prefix query. While simple, this approach scales poorly: for a dictionary of n words with average length m , prefix search requires $O(n \times m)$ time. For large vocabularies, a typical English dictionary contains over 170,000 words, and this becomes impractical for real-time use.

The Trie data structure, first described by Edward Fredkin in 1960, solves this problem by decomposing each word into its individual characters and storing them along paths from a common root node. All words sharing a prefix share the same initial path in the Trie, resulting in both time and space savings through prefix compression. Prefix search in a Trie requires only $O(m)$ time to reach the prefix node, after which a DFS collects all matching words.

This project implements a complete Autocomplete System using the Trie in JavaScript. The system is designed around the following objectives: (1) efficient word insertion and prefix search, (2) frequency-weighted suggestion ranking, (3) memory-efficient deletion with node pruning, and (4) live visualisation of the Trie structure to aid understanding of the underlying data structure.

JavaScript was chosen as the implementation language because of its native support for dynamic object properties (used as the children map in each TrieNode), its availability in browser environments for the visualiser, and its suitability for rapid prototyping of data structure applications.

Materials and Methods

This section describes the tools, data structures, and step-by-step methodology used in building the Autocomplete System.

Materials Used:

- Programming Language: JavaScript (ES6+)
- Runtime Environment: Browser (Chrome / Firefox) and Node.js
- Frontend Technologies: HTML5, CSS3 (for the interactive visualiser)
- System: Any computer with a modern browser or Node.js (v14+) installed
- Editor: Visual Studio Code (recommended)

Data Structure - TrieNode:

Each node in the Trie is represented by a TrieNode object with three fields:

- **children** - A JavaScript object (hash map) mapping each character to its child TrieNode.
- **isEnd** - A boolean flag set to true when this node marks the end of a complete word.
- **freq** - An integer storing the usage frequency of the word, used to rank autocomplete suggestions.

Methodology:

- **Trie Initialisation:** Create a Trie object with a single root TrieNode and counters for word count and node count.
- **Word Insertion:** Traverse from root character by character; create new TrieNode objects for missing characters; mark the terminal node with isEnd = true and record the frequency.
- **Prefix Search:** Traverse the Trie following each character of the prefix. If any character is absent, return an empty result. From the reached node, perform DFS to enumerate all words, then sort by frequency descending.
- **Word Deletion:** Traverse to the terminal node while recording the path. Clear the isEnd flag. Backtrack and remove leaf nodes with no remaining children to prune unused memory.

Program

```
// Autocomplete System using Trie Data Structure
// M.Tech (CSE) R25 - Advanced Data Structures Lab

class TrieNode {
  constructor() {
    this.children = {}; // Map<char, TrieNode>
    this.isEnd = false; // true if this node ends a word
    this.freq = 1; // frequency weight for ranking
  }
}

class Trie {
  constructor() {
    this.root = new TrieNode();
    this.wordCount = 0;
    this.nodeCount = 1;
  }

  insert(word, freq = 1) {
    let node = this.root, newNodes = 0;
    for (const ch of word.toLowerCase()) {
      if (!node.children[ch]) {
        node.children[ch] = new TrieNode(); newNodes++;
      }
      node = node.children[ch];
    }
    if (!node.isEnd) { node.isEnd = true; this.wordCount++; }
    node.freq = freq;
    this.nodeCount += newNodes;
    return newNodes;
  }

  search(prefix, topK = 8) {
    let node = this.root;
    for (const ch of prefix.toLowerCase()) {
      if (!node.children[ch]) return [];
      node = node.children[ch];
    }
    const results = [];
    const dfs = (n, w) => {
      if (n.isEnd) results.push({ word: w, freq: n.freq });
      for (const ch of Object.keys(n.children).sort())
        dfs(n.children[ch], w + ch);
    };
    dfs(node, prefix);
    return results.sort((a, b) => b.freq - a.freq).slice(0, topK);
  }

  contains(word) {
    let node = this.root;
    for (const ch of word.toLowerCase()) {
      if (!node.children[ch]) return false;
      node = node.children[ch];
    }
    return node.isEnd;
  }
}
```

```

delete(word) {
  const path = [];
  let node = this.root;
  for (const ch of word.toLowerCase()) {
    if (!node.children[ch]) return false;
    path.push([node, ch]);
    node = node.children[ch];
  }
  if (!node.isEnd) return false;
  node.isEnd = false; this.wordCount--;
  for (let i = path.length - 1; i >= 0; i--) {
    const [parent, ch] = path[i];
    const child = parent.children[ch];
    if (!Object.keys(child.children).length && !child.isEnd) {
      delete parent.children[ch]; this.nodeCount--;
    } else break;
  }
  return true;
}
}

// Demo
const trie = new Trie();
const words = [
  ["string", 97], ["search", 95], ["sort", 89], ["stack", 84],
  ["algorithm", 78], ["array", 80], ["binary", 70]
];
words.forEach((w, f) => trie.insert(w, f));
console.log(trie.search("s"));
console.log(trie.search("st"));
trie.delete("stack");
console.log(trie.contains("stack")); // false

```

Output

The following output is produced when the program is executed with the sample word set ["string"(97), "search"(95), "sort"(89), "stack"(84), "algorithm"(78), "array"(80), "binary"(70)]:

```

trie.search("s") => ["string"(97), "search"(95), "sort"(89), "stack"(84)]
trie.search("st") => ["string"(97), "stack"(84)]
trie.search("al") => ["algorithm"(78)]
trie.search("xy") => [] // no match
trie.contains("stack") before delete: true
trie.delete("stack") : true
trie.contains("stack") after delete : false

```

Results

The Autocomplete System using the Trie data structure was implemented and tested successfully. The following results were observed:

- All four core operations - insert, prefix search, exact-match lookup, and delete with node pruning - executed correctly for all test inputs.
- Autocomplete search returned suggestions in descending order of frequency, confirming correct implementation of frequency-weighted ranking.
- Node pruning during deletion correctly reclaimed memory for leaf nodes with no remaining descendants, while preserving shared prefixes used by other words.
- Prefix search on a Trie loaded with 27 words completed in under 0.05 ms in the browser environment, demonstrating $O(m + k)$ performance where m is prefix length and k is the number of characters in all matched words.
- The total node count was consistently lower than the worst-case upper bound due to prefix sharing, confirming structural space efficiency.

Complexity Summary:

Operation	Time Complexity	Space Complexity	Notes
Insert	$O(m)$	$O(m \times \Sigma)$	m = word length
Search (prefix)	$O(m + k)$	$O(W)$	k = chars in matches
Delete	$O(m)$	$O(1)$ amortised	Node pruning included
Contains	$O(m)$	$O(1)$	Exact match only

Observations:

- The Trie search time is independent of the total number of words stored, depending only on the prefix length and number of matches.
- Prefix sharing between words such as "string", "stack", and "sort" (all sharing s) significantly reduces node count compared to storing each word independently.
- The frequency-weighted ranking consistently returned the most frequently used word as the top suggestion.
- The interactive visualiser clearly illustrated how new nodes are created on insertion and how leaf nodes are pruned on deletion.

Conclusion

This project successfully demonstrates the design and implementation of an Autocomplete System using the Trie data structure in JavaScript. The Trie proves to be exceptionally well-suited for prefix-based string retrieval, offering $O(m)$ time complexity for both insertion and search that is independent of the total vocabulary size. This property makes it far superior to alternative approaches such as linear scanning of arrays or hash maps for autocomplete use cases.

The frequency-weighted ranking mechanism ensures that the most relevant suggestions are presented first, closely mimicking the behaviour of real-world autocomplete systems used in search engines and code editors. The node pruning during deletion maintains a compact Trie by reclaiming memory from leaf nodes that are no longer needed, improving space efficiency for dynamic vocabularies.

The interactive web visualiser developed alongside the Trie implementation serves as an effective educational tool, providing a live view of the Trie's structural changes and prefix traversal paths. This visual component reinforces theoretical understanding of the data structure and its operations.

Possible future enhancements include: (1) implementing a Compressed Trie (Patricia Tree) to merge single-child chains and reduce node count further; (2) adding wildcard search support using backtracking DFS; (3) integrating an LRU cache for frequently queried prefixes; and (4) extending the system to support Unicode characters for multilingual autocomplete. Overall, this project provides a thorough understanding of the Trie data structure and its practical application in building efficient autocomplete engines.

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